§1 Real-Rooted Polynomial Lower Bounds

This section treats the structural minimization of monic real-rooted polynomials over positive domains, utilizing classical inequalities and symmetric functional structures.

Problem Statement

Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + 1$ be a polynomial with positive real coefficients. If the roots of $P(x)$ are all real numbers, prove that:

$$P(2) \geq 3^n$$

Source: **Algebraic Extremal Problems**

1.1 Rigorous Algebraic Proof

Proof. Let the complete set of roots belonging to the polynomial $P(x)$ be denoted by $r_1, r_2, \ldots, r_n$.

We observe that all the coefficients a_k of the polynomial are explicitly given as positive real numbers. Because of this, substituting any positive real value into x will naturally result in a strictly positive sum. It is therefore impossible for $P(x)$ to possess any positive real roots. Since we are given that all n roots are entirely real, it follows that every root must be strictly negative.

To shift our domain to positive quantities, we can assign a positive value $x_i = -r_i$ for each index $i = 1, 2, \ldots, n$. Writing our polynomial in its factored form via the fundamental theorem of algebra gives:

$$P(x) = (x - r_1)(x - r_2) \ldots (x - r_n)$$

Substituting $r_i = -x_i$ into this factored expression reveals a product of binomial sums:

$$P(x) = (x + x_1)(x + x_2) \ldots (x + x_n)$$

We can analyze the constant term of our polynomial by evaluating it directly at $x = 0$. According to the problem definition, the trailing constant equals 1. Using Vieta's relations, this links the product of our transformed positive variables directly to unity:

$$P(0) = x_1 \cdot x_2 \ldots x_n = 1$$

Evaluating our expanded polynomial expression at the specific input value $x = 2$ gives:

$$P(2) = (2 + x_1)(2 + x_2) \ldots (2 + x_n)$$

We can find a lower bound for each individual term $(2 + x_i)$ by decomposing the constant 2 into a sum of ones and applying the Arithmetic Mean-Geometric Mean

(AM-GM) inequality across three positive terms ($1, 1$, and x_i):

$$\frac{1+1+x_i}{3} \geq \sqrt[3]{1 \cdot 1 \cdot x_i}$$

Simplifying this relation produces:

$$\frac{2+x_i}{3} \geq \sqrt[3]{x_i} \implies 2+x_i \geq 3\sqrt[3]{x_i}$$

Because each x_i is strictly positive, this minimum bound holds true independently for every factor in our product. Multiplying all n corresponding inequalities together allows us to construct a combined boundary for $P(2)$:

$$P(2) = \prod_{i=1}^{n}(2+x_i) \geq \prod_{i=1}^{n}\left(3\sqrt[3]{x_i}\right)$$

Factoring out the constants yields:

$$P(2) \geq 3^n \sqrt[3]{x_1 \cdot x_2 \ldots x_n}$$

We established earlier from the polynomial's constant term that the product of these elements $x_1 \cdot x_2 \ldots x_n$ equals exactly 1. Substituting this value back into our inequality simplifies our expression cleanly:

$$P(2) \geq 3^n \sqrt[3]{1}$$

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This completes the verification. The lower bound is achieved if and only if $x_i = 1$ for all i, corresponding to the specific polynomial $P(x) = (x+1)^n$. □